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CRIME ANALYTICS DASHBOARD

Insights Report

COURSE: BUSINESS INTELLIGENCE AND DATABASE MANAGEMENT SYSTEMS

Professor:
PROF. Ameni Azzouz

Prepared by:

Sarah Mneja
Maryem Ammar
Rissen Guermazi
Sirine Zaltni

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1. EXECUTIVE SUMMARY

This report presents analytical insights derived from a Business Intelligence (BI) dashboard developed using a star schema data warehouse and visualized in Power BI. The objective of the dashboard is to transform raw crime data into actionable insights that support decision-making related to public safety, law enforcement effectiveness, and resource allocation.

The analysis focuses on crime volume, temporal trends, geographic distribution, crime characteristics, and arrest outcomes. The findings reveal clear crime concentration patterns across districts and time periods, recurring seasonal and hourly trends, and variations in arrest performance. Based on these insights, the report proposes data-driven recommendations to enhance policing strategies and improve operational efficiency.

2. DASHBOARD MEASURES AND VISUALIZATIONS

The Power BI dashboard was designed around key analytical measures, each visualized using appropriate chart types to ensure clarity and interpretability.

2.1 CORE VOLUME MEASURES

Total Incident Count

- Description: Displays the total number of reported crime incidents.
- Purpose: Provides a high-level overview of overall crime volume.

Incidents per District

- Purpose: Identifies districts with the highest crime concentration.

Incidents per Community Area

- Description: Displays the total number of reported crime incidents.
- Purpose: Provides a high-level overview of overall crime volume.

2.2 TEMPORAL MEASURES

Monthly Incident Count

- Purpose: Shows how crime volumes change over time and identifies monthly trends.

Month-over-Month Trend

- Purpose: Measures growth or decline in incidents between consecutive months.

Peak Incident Hour

- Purpose: Identifies the hour of the day with the highest crime occurrence.

Seasonality Index

- Purpose: Examines crime variations across seasons (Winter, Spring, Summer, Fall).

2.3 INCIDENT CHARACTERISTICS

Incident Count by Primary Type

- Purpose: Identifies the most frequent crime categories.

Location-Type Distribution (%)

- Purpose: Shows where crimes most commonly occur (streets, residences, businesses).

2.4 OPERATIONAL & OUTCOME MEASURES

Weekday vs Weekend Incidents

- Purpose: Compares crime frequency across different days.

Daytime vs Nighttime

- Purpose: Identifies high-risk time windows.

Arrest Rate (%)

- Purpose: Measures law enforcement effectiveness.

Arrest Rate by District

- Purpose: Compares arrest success across districts.

Average Data Update

- Purpose: Evaluates data freshness and reliability.

3. KEY FINDINGS AND INSIGHTS

3.1 OVERALL CRIME VOLUME

The dashboard indicates a consistently high volume of reported crime incidents throughout the analyzed period. This suggests that crime is a persistent issue rather than a result of short-term fluctuations.

Why this matters:
Persistent crime levels require long-term strategies and sustained investment rather than reactive measures.

3.2 GEOGRAPHIC CRIME DISTRIBUTION

Crime incidents are unevenly distributed geographically. A limited number of districts and community areas account for a disproportionately high share of total incidents.

Why this matters:
Targeting high-risk areas can significantly improve the efficiency of policing efforts and resource allocation.

3.3 TEMPORAL PATTERN

Monthly analysis reveals noticeable fluctuations in crime volume, indicating seasonal effects. Additionally, crime occurrences peak during late-night hours, particularly around midnight.

Why this matters:
Understanding when crimes occur enables better patrol scheduling and proactive intervention during high-risk periods.

3.4 SEASONALITY EFFECTS

Crime volumes are higher during warmer seasons, especially in summer.

Why this matters:
Seasonal planning allows law enforcement agencies to anticipate increases in crime and allocate resources accordingly.

3.5 CRIME CHARACTERISTICS

A small number of crime types account for the majority of incidents.

Why this matters:
Targeted prevention programs focused on these dominant crime categories can produce more effective outcomes.

3.6 ARREST OUTCOMES

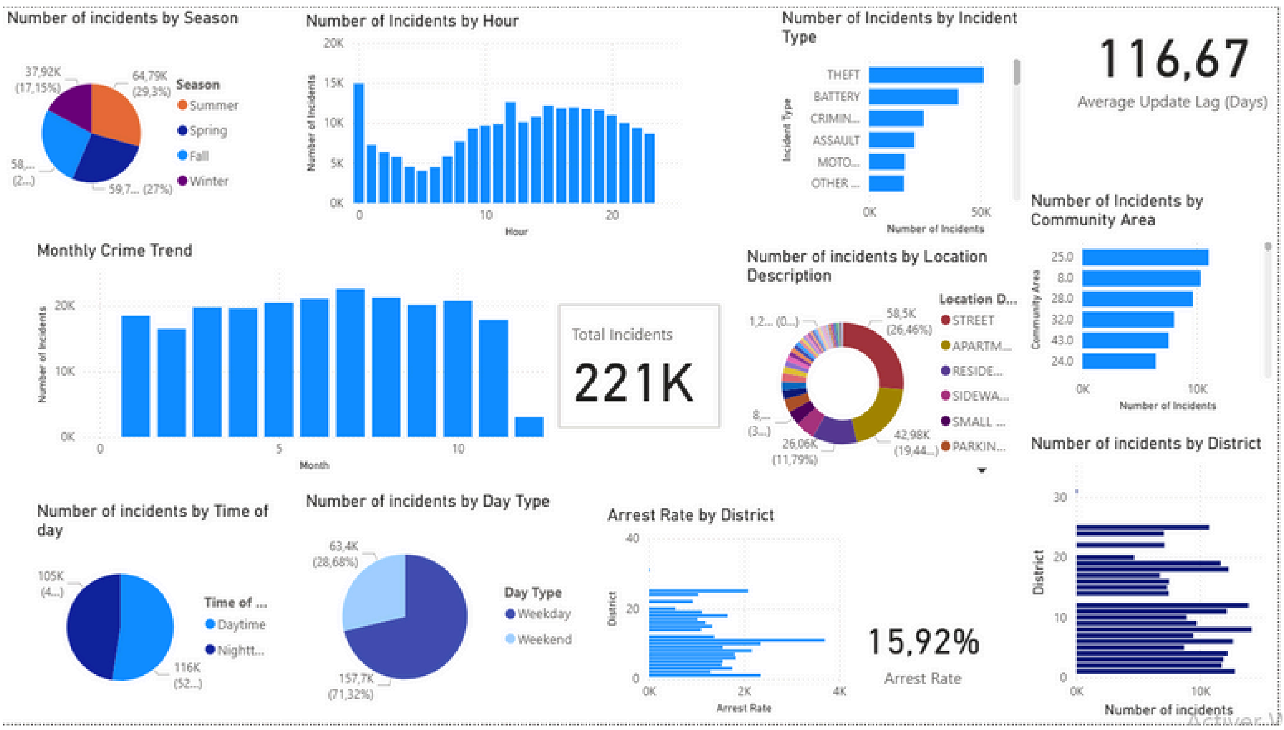
Arrest rates vary significantly by district and crime type.

Why this matters:
Differences in arrest rates may reflect variations in investigative complexity, operational efficiency, or resource availability.

3.7 DATA QUALITY

The average data update lag is relatively low, indicating timely data reporting.

Why this matters:
Reliable and up-to-date data ensures that analytical insights accurately reflect current crime conditions.



4. BUSINESS RECOMMENDATIONS

1. Implement Hotspot-Focused Policing

Concentrate patrols and surveillance efforts in districts and community areas with the highest incident rates.

2. Strengthen Nighttime Patrols

Increase police presence during late-night hours when crime risk is highest.

3. Adopt Seasonal Resource Planning

Allocate additional resources during high-crime seasons, particularly summer months.

4. Develop Targeted Crime Prevention Programs

Focus prevention strategies on the most common crime types identified in the dashboard.

5. Conduct District-Level Performance Reviews

Investigate districts with low arrest rates to identify operational challenges.

6. Use Dynamic Staffing Models

Adjust staffing levels based on weekday and weekend crime patterns.

6. Support Data-Driven Policy Design

Use dashboard insights to inform long-term public safety strategies.

5. LIMITATIONS AND FUTURE IMPROVEMENTS

5.1 Limitations

- The analysis relies on reported crime data and may not capture unreported incidents.
- Socio-economic and demographic factors were not included.
- The dashboard focuses on descriptive analytics rather than predictive modeling.
- Arrest rates do not fully reflect case resolution or judicial outcomes.

5.2 Future Improvements

- Integrate demographic and socio-economic datasets.
- Implement predictive analytics to forecast future crime trends.
- Enable real-time data ingestion for live monitoring.
- Expand outcome measures to include case resolution time and conviction rates.
- Enhance geospatial analysis for fine-grained hotspot detection.

6. CONCLUSION

The Crime Analytics Dashboard successfully transforms complex crime data into meaningful insights by integrating temporal, geographic, and operational dimensions. The analysis reveals clear crime patterns that can support evidence-based decision-making for law enforcement and policymakers. By addressing identified limitations and implementing the proposed recommendations, organizations can significantly enhance public safety outcomes and operational efficiency.

THANK YOU!